

Lab8db

May 14, 2026

1 Representations of the symmetric group

1.1 Math 737 - Lab 8

1.2 Exercise 1: For $n = 3, 4, 5, 6$, use the function ‘irreducible_characters()’ to show that the number of irreducible characters of S_n is the number of partitions of n .

```
[8]: S=[]
I=[]
for i in range(3,7):
    S = SymmetricGroup(i)
    print(f"It is {len(S.irreducible_characters())==Partitions(i)}.
↪cardinality()} that the number of partitions of {i} = the number of ↪
↪irreducible characters of S_{i}.")
```

It is True that the number of partitions of 3 = the number of irreducible characters of S_3 .

It is True that the number of partitions of 4 = the number of irreducible characters of S_4 .

It is True that the number of partitions of 5 = the number of irreducible characters of S_5 .

It is True that the number of partitions of 6 = the number of irreducible characters of S_6 .

1.3 Exercise 2: The code below does the following for $n = 3$. Do the same for $n = 4, 5$.

-

1.3.1 Compute the character table of S_n (including the conjugacy class representatives indexing the columns).

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1.3.2 By looking at the output and using your knowledge of character tables, state which partition corresponds to each row.

```
[ ]: S3 = SymmetricGroup(3)
display(S3.conjugacy_classes_representatives())
display(S3.character_table())
print("The partitions indexing the rows (in order) are:", [1,1,1], [2,1], [3])
```

```
[(), (1,2), (1,2,3)]
```

```
[ 1 -1  1]
```

```
[ 2  0 -1]
```

```
[ 1  1  1]
```

The partitions indexing the rows (in order) are: [1, 1, 1] [2, 1] [3]

```
[15]: S4 = SymmetricGroup(4)
display(S4.conjugacy_classes_representatives())
display(S4.character_table())
print("The partitions indexing the rows (in order) are:", [1,1,1,1], [2,1,1], [2,2], [3,1], [4])
```

```
[(), (1,2), (1,2)(3,4), (1,2,3), (1,2,3,4)]
```

```
[ 1 -1  1  1 -1]
```

```
[ 3 -1 -1  0  1]
```

```
[ 2  0  2 -1  0]
```

```
[ 3  1 -1  0 -1]
```

```
[ 1  1  1  1  1]
```

The partitions indexing the rows (in order) are: [1, 1, 1, 1] [2, 1, 1] [2, 2] [3, 1] [4]

```
[16]: S5 = SymmetricGroup(5)
display(S5.conjugacy_classes_representatives())
display(S5.character_table())
print("The partitions indexing the rows (in order) are:", [1,1,1,1,1], [2,1,1,1], [2,2,1], [3,1,1], [3,2], [4,1], [5])
```

```
[(), (1,2), (1,2)(3,4), (1,2,3), (1,2,3)(4,5), (1,2,3,4), (1,2,3,4,5)]
```

```
[ 1 -1  1  1 -1 -1  1]
```

```
[ 4 -2  0  1  1  0 -1]
```

```
[ 5 -1  1 -1 -1  1  0]
```

```
[ 6  0 -2  0  0  0  1]
```

```
[ 5  1  1 -1  1 -1  0]
```

```
[ 4  2  0  1 -1  0 -1]
```

```
[ 1  1  1  1  1  1  1]
```

The partitions indexing the rows (in order) are: [1, 1, 1, 1, 1] [2, 1, 1, 1] [2, 2, 1] [3, 1, 1] [3, 2] [4, 1] [5]

1.4 Exercise 3: The code below does the following for $n = 3$. Do the same for $n = 4, 5$.

•

1.4.1 Check that the row orthogonality theorem holds.

```
[ ]: irr = S3.irreducible_characters()
      #irr is the list of irreducible characters. Since it is a list, we can access
      ↪ each of these three elements as irr[0], irr[1], irr[2].
      display(irr[0].values())
      display(irr[1].values())
      display(irr[2].values())
```

[1, -1, 1]

[2, 0, -1]

[1, 1, 1]

```
[21]: #Now use .scalar_product to perform the inner product of one character with
      ↪ another.
      display(irr[0].scalar_product(irr[0]))
      display(irr[0].scalar_product(irr[1]))
      display(irr[0].scalar_product(irr[2]))

      display(irr[1].scalar_product(irr[0]))
      display(irr[1].scalar_product(irr[1]))
      display(irr[1].scalar_product(irr[2]))

      display(irr[2].scalar_product(irr[0]))
      display(irr[2].scalar_product(irr[1]))
      display(irr[2].scalar_product(irr[2]))
      #Need to check the scalar product for all combinations of irr[i], irr[j] for
      ↪ values of i,j between 0 and 2. You'll want to write a loop for higher n,
      ↪ since there will be many more combinations for S_4, S_5, and S_6.
```

1

0

0

0

1

0

0

0

1

For S_4, S_5 I show it in matrix form where $M = (\langle \chi_i, \chi_j \rangle) = (\delta(i, j)) = I$

S_4

```
[31]: irr = S4.irreducible_characters()

for i in range(len(irr)):
    display(irr[i].values())

print("-----")

M = matrix([
    [irr[i].scalar_product(irr[j]) for j in range(len(irr))]
    for i in range(len(irr))
])

display(M)
```

[1, -1, 1, 1, -1]

[3, -1, -1, 0, 1]

[2, 0, 2, -1, 0]

[3, 1, -1, 0, -1]

[1, 1, 1, 1, 1]

[1 0 0 0 0]

[0 1 0 0 0]

[0 0 1 0 0]

[0 0 0 1 0]

[0 0 0 0 1]

S_5

```
[32]: irr = S5.irreducible_characters()

for i in range(len(irr)):
    display(irr[i].values())

print("-----")

M = matrix([
    [irr[i].scalar_product(irr[j]) for j in range(len(irr))]
    for i in range(len(irr))
])
```

```
])
```

```
display(M)
```

```
[1, -1, 1, 1, -1, -1, 1]
```

```
[4, -2, 0, 1, 1, 0, -1]
```

```
[5, -1, 1, -1, -1, 1, 0]
```

```
[6, 0, -2, 0, 0, 0, 1]
```

```
[5, 1, 1, -1, 1, -1, 0]
```

```
[4, 2, 0, 1, -1, 0, -1]
```

```
[1, 1, 1, 1, 1, 1, 1]
```

```
[1 0 0 0 0 0 0]
```

```
[0 1 0 0 0 0 0]
```

```
[0 0 1 0 0 0 0]
```

```
[0 0 0 1 0 0 0]
```

```
[0 0 0 0 1 0 0]
```

```
[0 0 0 0 0 1 0]
```

```
[0 0 0 0 0 0 1]
```

1.5 Exercise 4: The code below does the following for $n = 3$. Do the same for $n = 4, 5$.

•

1.5.1 Sum the squares of the dimensions of each irreducible representation to obtain the order of the group.

```
[39]: #Here is the calculation for n=3. For higher n, you should write a loop rather
      ↪ than writing the sum manually.
```

```
irr = S3.irreducible_characters()
list(irr[0])[0]^2+list(irr[1])[0]^2+list(irr[2])[0]^2
```

```
[39]: 6
```

The below cells show that we get $|S_4| = 24 = 4!$, $|S_5| = 5! = 120$, respectively, as expected, when summing the squares of the dimensions of each irreducible representation.

```
[40]: S4 = SymmetricGroup(4)
      irr = S4.irreducible_characters()

      sum(list(irr[i])[0]^2 for i in range(len(irr)))
```

```
[40]: 24
```

```
[41]: S5 = SymmetricGroup(5)
      irr = S5.irreducible_characters()

      sum(list(irr[i])[0]^2 for i in range(len(irr)))
```

```
[41]: 120
```